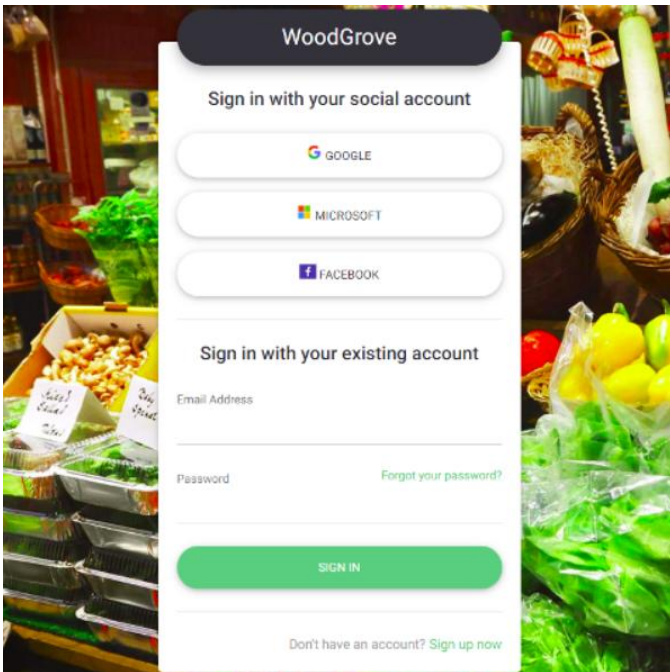




Azure Active
Directory B2C

Day 11 Azure Learning Series

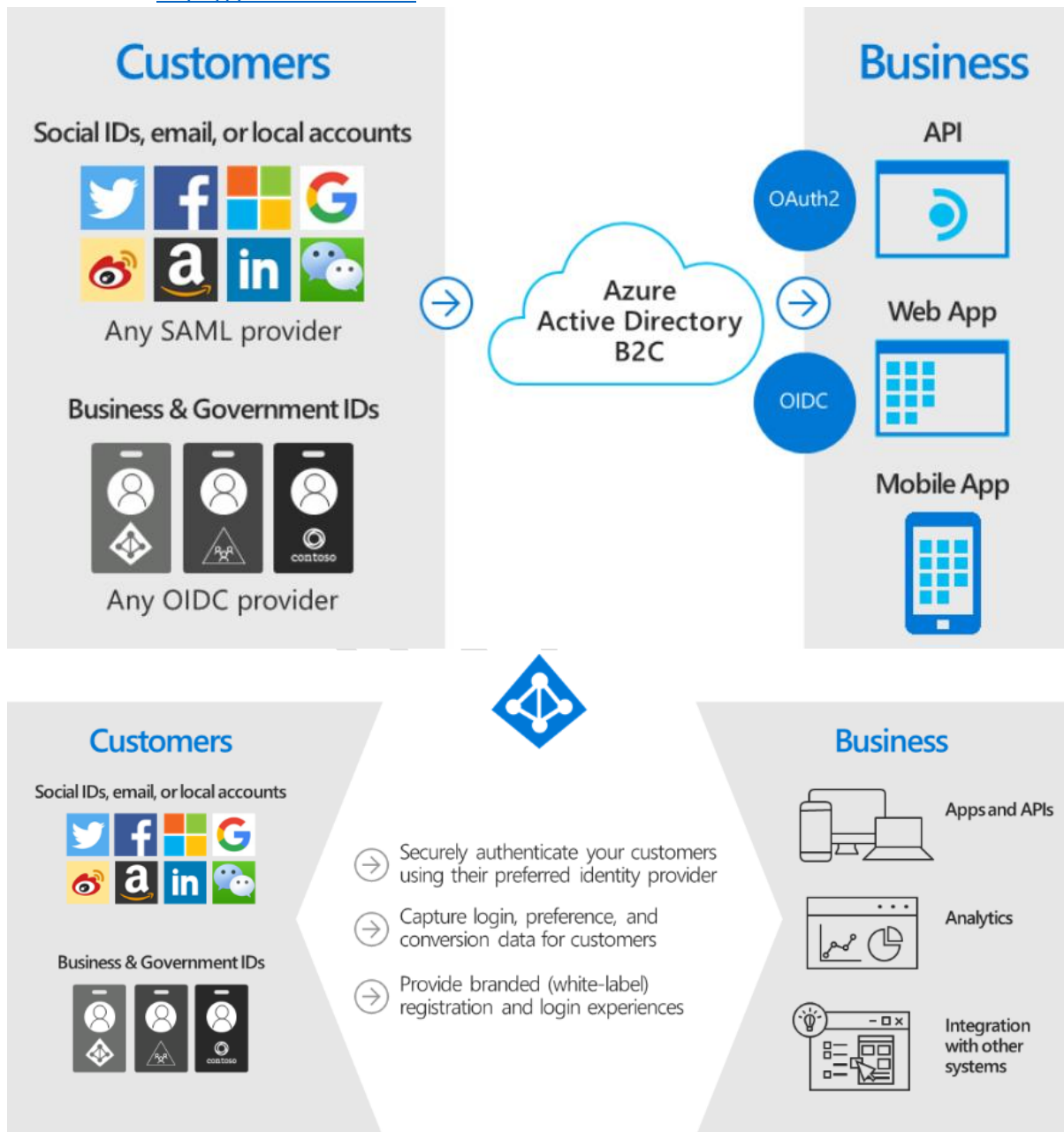
With Sai Reddy

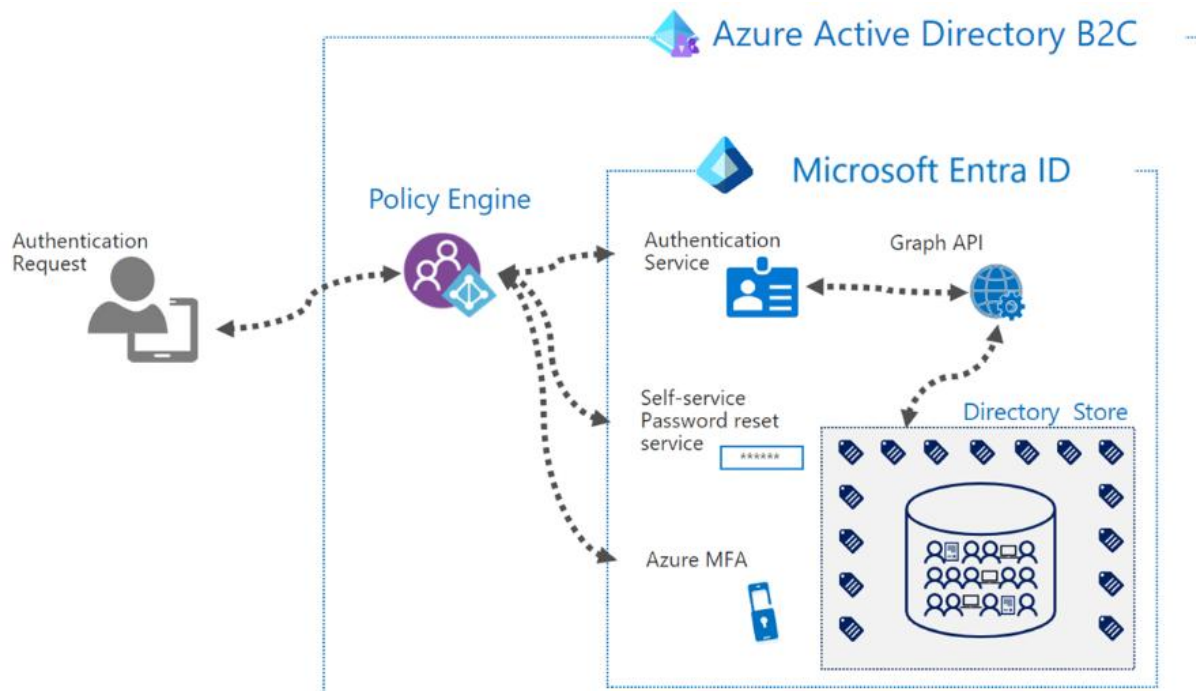
Day 11 Azure Learning Series

Azure AD B2C

Step 1: Log into Azure Portal

- Go to <https://portal.azure.com>





What is Azure AD B2C?

Azure Active Directory B2C is an identity and access management service that enables your applications to authenticate customers using social identities (Google, Facebook, etc.) or local accounts (email/password) in a secure, scalable way.

When Should You Use Azure AD B2C?

- You have **consumer-facing apps** (web or mobile).
- You need **SSO** across multiple apps for customers.
- You want to allow **login with social identities** (e.g., Google, Facebook, GitHub).
- You want to **customize** the authentication experience (branding, layout, flow).
- You want a **centralized identity management system** without managing user databases.

Step-by-Step: Setup Azure AD B2C and Integrate with Angular App

Step 1: Create Azure AD B2C Tenant

1. Go to [Azure Portal](#)
2. Search for **Azure AD B2C** → Click **Create a new Azure AD B2C Tenant**
3. Provide:
 - Organization name
 - Initial domain name (e.g., myb2ctenant.onmicrosoft.com)
 - Country/Region
4. After creation, go back to Azure Portal → Switch to the **new B2C directory**.

Step 2: Register Your Angular App in B2C

1. In B2C directory → Go to App registrations → Click + New registration
2. Enter:
 - Name: MyAngularApp
 - Redirect URI: http://localhost:4200 (for dev)
 - Choose Single-page application (SPA)
3. After registration:
 - Copy the Application (client) ID
 - Go to Authentication tab → Check ID tokens (used for implicit/hybrid flow)

Step 3: Create User Flows (Policies)

1. Go to **User flows** → Click + **New user flow**
2. Choose:
 - **Sign up and sign in**
 - (Optional) Password reset, Profile edit
3. Choose identity providers (Microsoft, Facebook, Google, local, etc.)
4. Name the flow: B2C_1_signupsigin
5. Click **Create**

Step 4: Configure Identity Providers (Optional)

- For Google/Facebook login: Configure API keys in Identity providers section

Azure AD B2C is charged per authentication:

Tier	Monthly Authentications	Cost
Free	First 50,000	Free
Standard	50,001–100,000	~\$0.0028 per auth
Custom	Advanced scenarios	Extra charges for custom policies, MFA, etc.

Example: 100,000 monthly logins ≈ \$140/month

[Detailed Pricing](#)

Advantages

- Secure authentication (OAuth2, OpenID Connect)
- SSO across multiple apps
- Supports social + local identities
- Customizable UI and branding
- Highly scalable and managed by Microsoft
- Token-based auth (JWT) for SPAs and APIs

Disadvantages

- Initial setup can be complex for custom flows
- Limited out-of-the-box customization for UI without custom policies
- Can get expensive with high authentication volumes
- Debugging errors may require Azure logs

[Home](#) > [Default Directory | Overview](#) >

Manage tenants ...

[+](#) Create [↻](#) Refresh [≡](#) Columns | [↔](#) Switch [🗑](#)

Current tenant: Default Directory

[+ Add filters](#)

Showing 1 of 1 results

☐ Organization name ↑↓

☐  Default Directory (Default)

Microsoft Azure

Upgrade

Search resou

[Home](#) >
[Default Directory | Overview](#) >
[Manage tenants](#) >

Create a tenant

* Basics

* Configuration

Review + create

Microsoft Entra ID and Azure AD B2C enable users to access applications published by your orga

Tenant type

Customers must own a paid license to create Microsoft Entra Workforce tenant.

As of May 1 2025, Azure AD B2C is no longer available for new sales. If you are an existing custor

Select a tenant type *

☐ Microsoft Entra ID

☒ Azure AD B2C

☐ Microsoft Entra External ID

[Help me choose...](#)

Review + create

< Previous

Next : Configuration >

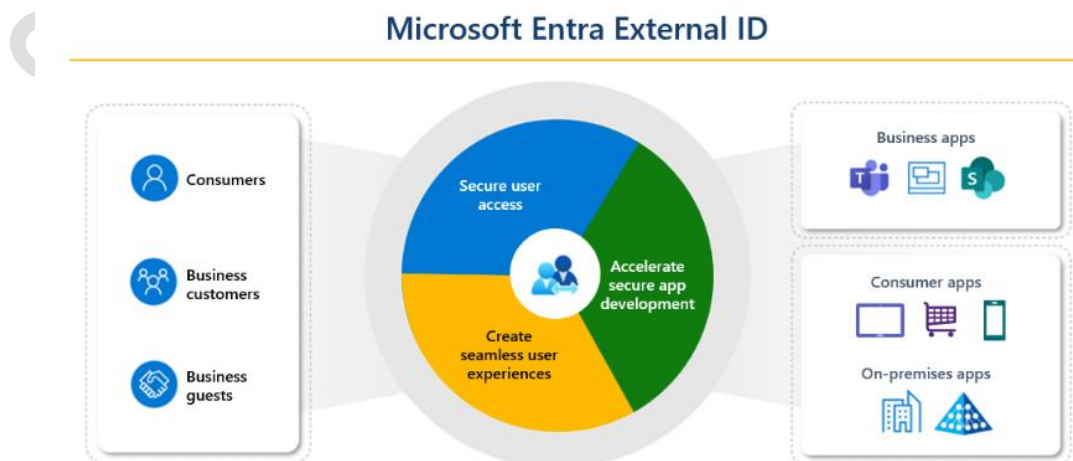
Note : as of May 1, 2025, Azure AD B2C is deprecated for new tenants and no longer available for new sales or new tenant creation. While creating Azure AD B2C I have seen this error. Then I proceeded with **Microsoft Entra External ID**.

What is Microsoft Entra External ID (CIAM for Customers)?

It's a cloud-based customer identity and access management (CIAM) service.

It helps you:

- Authenticate and manage external users (customers, partners)
- Provide user sign-up, sign-in, and self-service profile management
- Secure APIs and apps using OpenID Connect, OAuth2, and MSAL



Why Azure AD B2C is Being Replaced?

- B2C had limitations in extensibility, usability, and modern security protocols.
- Entra External ID provides a more unified, modern, and scalable CIAM solution with support for:
 - Built-in user flows (Sign up, sign in, password reset)
 - Custom attributes
 - External identity providers (Google, Facebook, etc.)
 - Custom domains
 - Built-in templates and policies
 - Integration with Microsoft Graph & conditional access

Pricing: Entra External ID (As of 2025)

Entra External ID offers a **consumption-based pricing model**:

Free Tier:

- **50,000 monthly active users (MAUs)**
- Includes core features like user flows, password reset, multifactor auth (MFA), SSO, etc.
- Ideal for startups or MVPs.

Paid Tier:

- After 50,000 MAUs: Around **\$0.00325 per MAU/month**
- Add-ons (optional):
 - Premium Conditional Access
 - Identity protection
 - Advanced security reports

MAU = Monthly Active User

A unique user who signs in at least once in a month counts as 1 MAU.

You're only billed for users who log in during a month, not for total registered accounts.

Full pricing page: <https://azure.microsoft.com/en-us/pricing/details/entra/external-id/>

When Should You Use Entra External ID?

Use it when:

- You are building an app used by external customers (e.g., end users, consumers, citizens).
- You need custom sign-up/sign-in with MFA, password reset, etc.
- You want social logins (Google, Facebook, Apple, etc.).
- You want enterprise-grade security + extensibility.

When Not To Use It

Avoid using Entra External ID if:

- You're only managing internal users/employees (use **Entra ID / Azure AD** instead).
- You want full on-premise identity management (use something like ADFS or a hybrid setup).

Create a tenant ...

Directory details

Configure your new directory

Organization name *	MyB2CApp
Initial domain name *	saireddydotnetfs
Country/Region	India
Select an Azure subscription and resource group	A resource group is a container that holds related resources for an Azure solution. Name * myb2cResourcegroup OK Cancel
Subscription *	
Resource group *	

[Create new](#)

[Review + create](#)

[< Previous](#)

[Next : Review + create >](#)

[Home](#) > [Default Directory | Overview](#) > [Manage tenants](#) >

Create a tenant ...

✓ Validation passed.

Summary

Basics

Tenant type Azure AD B2C

Configuration

Organization name	MyB2CApp
Initial domain name	saireddydotnetfs.onmicrosof
Country/Region	India
Subscription	Azure subscription 1
Resource group	myb2cResourcegroup
Resource group location	East US

[Create](#)

[< Previous](#)

[Next >](#)

Home >

myExternalEntraIdTenant | Overview ...





 Add

 Manage tenants

 What's new

 Overview

 Preview features

 Diagnose and solve problems



 Users

 Groups

 External Identities

 Roles and administrators

 Delegated admin partners

 Enterprise applications

 Devices

 App registrations

 Identity Governance

 Application proxy

 Custom security attributes

 Licenses

 [Microsoft Entra has a simpler, integrated experience](#)

Get started

Overview

Properties



Try the get started guide

Switch to your Entra External ID tenant

Navigate to:

Home > Identity > Applications > App registrations

Click + New registration

Sai Reddy
saireddy-dotnetfs

All services >

App registrations

+ New registration Endpoints Troubleshoot Refresh Download Preview features Got feedback?

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

All applications Owned applications Deleted applications Applications from personal account

Start typing a display name or application (client) ID to filter these results

Add filters

All services > App registrations >

Register an application

myAngular18

Supported account types

Who can use this application or access this API?

- ☒ Accounts in this organizational directory only (myExternalEntraID)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID accounts (e.g. Skype, Xbox))
- ☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating and it can be changed later, but a value is required for most

Select a platform

http://localhost:4200

Public client/native (mobile & desktop)

Web

Single-page application (SPA)

Integrate gallery apps and other

[Platform Policies](#)

Register

All services > App registrations >

myAngular18

Search

Delete Endpoints Preview features

Overview

- Quickstart
- Integration assistant
- Diagnose and solve problems
- Manage
- Support + Troubleshooting

Essentials

Display name : myAngular18

Application (client) ID : f3c4ea3a-b455-42bd-a09e-0bbca23eb63d

Object ID : 1c68ca63-1d43-4dbc-a321-98aa7597aeecc

Directory (tenant) ID : f5130957-4437-42a7-8b70-0d0f6f728329

Supported account types : My organization only

Client credentials : [Add a certificate or secret](#)

Redirect URIs : [0 web, 1 spa, 0 public client](#)

Application ID URI : [Add an Application ID URI](#)

Managed application in ... : myAngular18

State : ☒ Activated

Create application
Successfully

Our Application has been created.

[All services](#) > [App registrations](#) > [myAngular18](#)

myAngular18 | Authentication ...

 Got feedback?

- Overview
- Quickstart
- Integration assistant
- Diagnose and solve problems
- Manage
 - Branding & properties
 - Authentication**
 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
 - App roles
 - Owners
 - Roles and administrators
 - Manifest
- Support + Troubleshooting

Platform configurations

Depending on the platform or device this application is targeting, additional configuration may be required such as redirect URIs, specific authentication settings, or fields specific to the platform.

[+ Add a platform](#)

Single-page application

[Quickstart](#) [Docs](#) 

Redirect URIs

The URIs we will accept as destinations when returning authentication responses (tokens) after successfully authenticating or signing out users. The redirect URI you send in the request to the login server should match one listed here. Also referred to as reply URLs. [Learn more about Redirect URIs and their restrictions](#)

[http://localhost:4200](#) 

[Add URI](#)

Grant types

 Your Redirect URI is eligible for the Authorization Code Flow with PKCE.

[Save](#) [Discard](#)

Enable ID Tokens

[All services](#) > [App registrations](#) > [myAngular18](#)

myAngular18 | Authentication ...

 Got feedback?

- Overview
- Quickstart
- Integration assistant
- Diagnose and solve problems
- Manage
 - Branding & properties
 - Authentication**
 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
 - App roles
 - Owners
 - Roles and administrators
 - Manifest
- Support + Troubleshooting

Select the tokens you would like to be issued by the authorization endpoint:

- ☐ Access tokens (used for implicit flows)
- ☒ ID tokens (used for implicit and hybrid flows)

Supported account types

Who can use this application or access this API?

- ☒ Accounts in this organizational directory only (myExternalEntraIdTenant only - Single tenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)

[Help me decide...](#)

Advanced settings

Allow public client flows

Enable the following mobile and desktop flows:

[Yes](#) [No](#)

- App collects plaintext password (Resource Owner Password Credential Flow) [Learn more](#)
- No keyboard (Device Code Flow) [Learn more](#)
- SSO for domain-joined Windows (Windows Integrated Auth Flow) [Learn more](#)

Enable native authentication

[Yes](#) [No](#)

[Save](#) [Discard](#)

Step 1: Create a User Flow (Signup/Signin Page)

This lets users register and login.

- Go to: <https://entra.microsoft.com>
- In the left menu, go to:
Identity → External Identities → User flows
- Click + New user flow
- Select "Sign up and sign in"
- Name it: SignupSignInFlow

Sai Reddy
saireddy-dotnetfs

6. Select Email with password (local account) ☒
7. Click Create

  **Microsoft Azure**

[All services >](#)

 **External Identities | Overview** 
myExternalEntralIdTenant

   Got feedback?

 **Overview**

 Cross-tenant access settings

 All identity providers

 External collaboration settings

 Diagnose and solve problems

 > Self-service sign up

 > Subscriptions

 > Lifecycle management

 > Troubleshooting + Support

Get started 

Get started with External Identities

External Identities is a set of capabilities that enable users, including customers and partners beyond your organization, to interact and connect with users outside your organization. [Learn more](#) 

[All services >](#) [External Identities](#)

 External Identities | User flows
myExternalEntralIdTenant

   **New user flow** |  Got feedback?

✓ Self-service sign up

 **User flows**

User flow name

Name

No user flows found.

Create

Sign up and sign in

Got a second? We would love your feedback on creating user

☒ Email Accounts

☒ Email with password

☐ Email one-time passcode

3. User attributes

User attributes are values collected on sign up. Claims are values

Collect attribute

Given Name ☐

Surname ☐

City ☐

Country/Region ☐

Display Name ☒

Also selected: Email Address

[Show more...](#)

Create

 Create user flow

Successfully created user flow SaiReddy.

Search for Microsoft Entra ID then Navigate to Users

Microsoft Azure

Search resources, services, and docs (G+)

Home > myExternalEntraIdTenant | Users >

Users

myExternalEntraIdTenant

< + New user Edit Delete Download users Bulk operations Refresh Manage view

☒ All users [Azure Active Directory is now Microsoft Entra ID.](#)

Search

1 user found

☐ Display name ↑	User principal name ↑↓	User type	On-premises sy...
---	------------------------	-----------	-------------------

[Home](#) >



Users

...

myExternalEntraIdTenant



New user ▾



Edit ▾

Delete



All users



Audit logs



Sign-in logs



Diagnose and solve problems



Deleted users



Password reset



User settings

Create new user

Create a new internal user in your organization

Invite external user (Preview)

Invite an external user to collaborate with your organization

Create new external user

Create a new customer, citizen, or business partner in your organization

[Home](#) > [Users](#) >

Create new external user

...

Create a new customer, citizen, or business partner in your organization

[Basics](#) [Properties](#) [Assignments](#) [Review + create](#)

Create a new user in your external tenant configuration. This user can sign in to your external tenant configuration with

Identity

Identities *

Sign-in method

Value

Email ▾

testsaireddy1@saireddydotnetfs.on...



+ Add

Display name *

Sai Reddy

Password *



☒ Auto-generate password

Account enabled ⓘ

☒

[Review + create](#)

[< Previous](#)

[Next: Properties >](#)

Create new external user ...

Create a new customer, citizen, or business partner in your organization

Basics Properties Assignments Review + create

Basics

Identities testsaireddy1@saireddydotnetfs.onmicrosoft.com
Display name Sai Reddy
Password
Account enabled Yes

Properties

First name Sai Reddy
User type Member

Assignments

Roles

Create < Previous Next >

Home >

Users ...
myExternalEntralTenant

+ New user Edit Delete Download users Bulk operations Refre

All users

Audit logs

Sign-in logs

Diagnose and solve problems

Deleted users

Password reset

Azure Active Directory is now Microsoft Entra ID.

SaiReddy

1 user found

Display name	User principal name	User type
SaiReddy	10055b3c-f48f-4d...	Member

Welcome to Our App

Please sign in using your Microsoft account.

Login with Microsoft

Logout

After clicking this Login with Microsoft



Sign in

TestSai@saireddydotnetfs.onmicrosoft.com

[Can't access your account?](#)

Next



testsai@saireddydotnetfs.onmicrosoft.com

Update your password

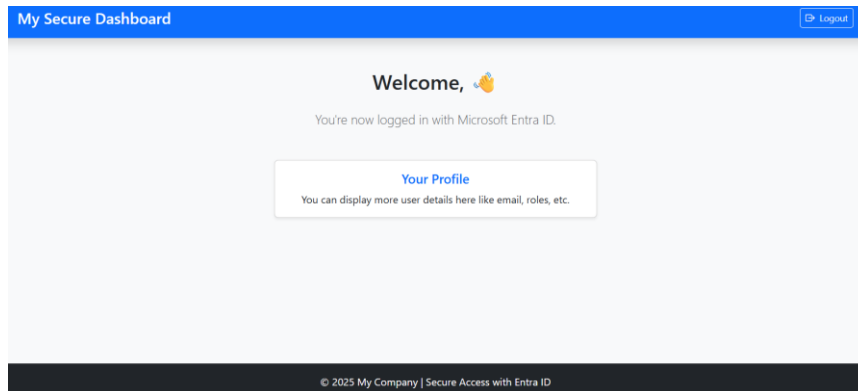
You need to update your password because this is the first time you are signing in, or because your password has expired.

.....

New password

Confirm password

Sign in



I have to Make part 2 for this document....

[Check full code in GitHub repository](#)

